

Real-Time Phishing Website Detection via Mutual Information-Driven Feature Selection and Random Forest Ensemble Classification

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Abstract—Phishing attacks represent one of the most prevalent and economically damaging threats in contemporary cybersecurity, exploiting counterfeit websites to harvest sensitive user credentials. This paper introduces a machine learning-based phishing website detection framework constructed upon the PhiUSIIL Phishing URL Dataset, encompassing 235,795 labelled URL samples. The original dataset comprises 56 features derived from URL structure, HTML content, and webpage metadata. To enhance model efficiency and reduce computational overhead, a feature selection methodology grounded in Mutual Information (MI) scoring was applied, contracting the feature space from 56 to 20 URL-extractable features with negligible performance degradation. Four machine learning algorithms were systematically evaluated: Random Forest, Decision Tree, Gradient Boosting, and Logistic Regression. The Random Forest classifier configured with 200 estimators delivered superior performance, attaining an accuracy of 97.38%, an AUC-ROC of 0.9973, and robust generalisation through 5-fold cross-validation yielding a mean accuracy of $97.36\% \pm 0.04\%$. A deterministic rule-based override layer was further incorporated to manage unambiguous phishing or legitimate signals with high confidence. The complete system is deployed as an interactive Streamlit web application enabling real-time URL classification. These findings affirm that a compact suite of URL-based features, paired with a robust ensemble classifier, yields an effective and practically deployable phishing detection solution.

Index Terms—Phishing detection, machine learning, Random Forest, mutual information, feature selection, URL analysis, cybersecurity, PhiUSIIL dataset

I. INTRODUCTION

Phishing is a form of social engineering cyberattack wherein adversaries craft fraudulent websites that closely replicate trusted entities, deceiving users into surrendering sensitive information such as login credentials, banking details, and personal identification data [1]. As global dependence on e-commerce, digital banking, and online communication continues to deepen, the frequency and sophistication of phishing attacks have grown correspondingly, posing substantial risks to individuals, enterprises, and public institutions [2]. Phishing detection has been an active research problem for nearly two decades, with early works such as Garera et al. [18] and Fette et al. [19] establishing foundational detection frameworks for phishing websites and emails respectively, and comprehensive surveys by Khonji et al. [23] systematically cataloguing the evolution of detection methodologies.

According to the Anti-Phishing Working Group, the volume of unique phishing websites reported quarterly escalated dramatically from approximately 164,772 in Q1 2020 to over 1.6 million by Q1 2023, underscoring the urgency of effective automated detection mechanisms [3]. Attackers continuously refine their strategies, employing domain spoofing, URL obfuscation, IP-based redirection, and low-reputation top-level domains (TLDs) to circumvent conventional rule-based and blacklist-based defences [4]. Large-scale automated classification approaches, such as the system proposed by Whittaker et al. [22], demonstrated that machine learning can scale to millions of webpages for phishing identification, highlighting the need for efficient and deployable detection pipelines.

Machine learning (ML) offers a compelling paradigm for phishing detection, enabling discriminative pattern learning from large labelled datasets without the need for manual rule engineering [5]. URL-based features are particularly well-suited for automated detection, as they can be derived directly from the URL string without requiring webpage retrieval, thus facilitating lightweight and real-time inference pipelines [6]. However, many phishing research datasets incorporate HTML and page-level features that are

computationally expensive to extract and impractical in deployment contexts where web crawling is infeasible or restricted [7].

This work addresses these challenges by leveraging the PhiUSIIL Phishing URL Dataset, which contains 235,795 URL instances with 56 features spanning both URL-level and HTML-level attributes. Mutual Information-based feature selection is applied to isolate a reduced set of 20 URL-extractable features, yielding a system that is simultaneously accurate and practical for real-world deployment. Multiple ML classifiers are compared, with the Random Forest algorithm emerging as the optimal model. The system is further augmented by a deterministic rule-based override layer and packaged as a Streamlit web application for interactive use.

This paper is structured as follows. Section II surveys the related literature. Section III describes the dataset. Section IV details the feature selection process. Section V presents the methodology. Section VI discusses experimental results. Section VII provides the conclusion and directions for future research.

II. LITERATURE SURVEY

A substantial body of research has explored machine learning approaches for phishing website detection, spanning URL-based, HTML-based, and hybrid methodologies. Early investigations primarily concentrated on lexical and URL-structural features. Jain and Gupta [16] proposed a machine learning approach leveraging hyperlink information for phishing detection, demonstrating the discriminative power of URL-structural signals. Mohammad et al. [17] introduced a self-structuring neural network for phishing prediction, establishing deep learning as a viable direction for URL-based classification. Abdelhamid et al. [24] explored associative classification techniques for phishing detection, achieving competitive results through rule-based pattern mining on URL and page-level attributes. Building on these foundational efforts, Sánchez-Paniagua et al. [15] introduced a novel multipurpose phishing dataset incorporating web technology features and demonstrated that integrating URL attributes with page-level signals can achieve accuracies exceeding 97% using tree-based classifiers.

Feature selection and dimensionality reduction have emerged as recurrent themes in recent literature. Abu Elsoud et al. [3] proposed a backward recursive feature selection framework that systematically prunes redundant attributes from phishing datasets, reporting improved classification accuracy alongside a substantial reduction in the feature count. Similarly, Liu et al. [13] introduced the CASE feature framework a multistage model that categorises website attributes into Content, Address, Structure, and External components achieving a false positive rate below 1% in live web environments.

Ensemble and deep learning methodologies have consistently demonstrated superior performance. Liu et al. [14] proposed multiscale semantic deep fusion models combining convolutional and recurrent architectures for phishing URL classification. Henry et al. [4] extended this paradigm through ensemble transfer learning, applying cross-dataset generalisation strategies to address distributional discrepancies between training and deployment data. Yeboah and Kayes [5] investigated self-supervised pre-trained models as feature extractors within an ensemble framework, achieving competitive results on established benchmarks.

Recent investigations have explored the intersection of large language models (LLMs) and phishing. Sivaneswaran et al. [1] conducted a systematic literature review documenting how LLMs are being weaponised to generate more persuasive phishing content, while concurrently noting emerging LLM-based detection approaches. Li et al. [2] analysed adversarial asymmetry in LLM-based phishing detection, establishing that semantic plausibility introduced by LLMs substantially increases the difficulty for automated classifiers.

Specialised phishing variants have also received considerable scholarly attention. Himdi and Alhayan [8] investigated Arabic-language phishing email detection using optimised ensemble stacking techniques. Yalda et al. [9] proposed a blockchain-based QR code verification framework to combat QR phishing attacks. Erfan et al. [6] examined phishing from a domain ownership perspective, providing longitudinal analysis of how phishing campaigns exploit borrowed domain infrastructure. Alzaylaee et al. [10] contributed a broader cybersecurity perspective through AI-driven computer vision and ML models tailored for real-time threat detection.

Graph-based and deep neural approaches have further enriched the landscape. Tchakounte et al. [12] developed Crawl-shing, a focused crawler employing graph isomorphism to retrieve phishing content, demonstrating the utility of structural URL representations. Crisostomo et al. [11] highlighted the broader applicability of Random Forest and ensemble methods in financial fraud detection, reinforcing the transferability of such approaches to phishing classification tasks. Furthermore, graph-based representation learning methods such as DeepWalk [25], originally developed for social network embeddings, have inspired analogous approaches in cybersecurity for learning structural representations of URL graphs and web topology, offering a promising direction for future phishing detection research.

The present work builds upon these contributions by combining systematic MI-based feature reduction with a Random Forest ensemble evaluated on the large-scale PhiUSIIL dataset, yielding a lightweight yet highly accurate detection system validated through rigorous cross-validation and multi-metric performance assessment.

III. DATASET

A. Dataset Description

The PhiUSIIL Phishing URL Dataset [University of California, Irvine, 2023] was employed in this study. The dataset encompasses 235,795 URL samples distributed across two classes: 134,850 legitimate URLs (label = 1) and 100,945 phishing URLs (label = 0),

representing an approximately 57:43 class ratio. Data collection involved aggregating URLs from reputable phishing repositories and legitimate web sources, ensuring diversity across attack typologies and legitimate domain categories.

Each sample is characterised by 56 features grouped into three broad categories: (1) URL structure features, extracted directly from the URL string, including URL length, domain length, subdomain count, HTTPS presence, TLD-based legitimacy probability, special character counts, and obfuscation ratio; (2) HTML and source code features requiring webpage retrieval, encompassing image count, CSS files, JavaScript references, iframe elements, external form submissions, and lines of code; and (3) metadata and similarity features, including URL-to-title match scores, domain title match scores, copyright information presence, and password field detection.

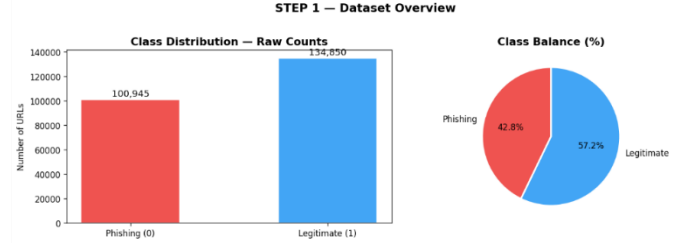


TABLE I
OVERVIEW OF THE PHIUSIIL DATASET

Property	Value	Notes
Total samples	235,795	Balanced representation
Legitimate URLs (label=1)	134,850	57.19%
Phishing URLs (label=0)	100,945	42.81%
Total features	56	Before reduction
Missing values	0	Complete dataset
Train split (80%)	188,636	Stratified
Test split (20%)	47,159	Stratified

B. Data Preprocessing

The dataset required minimal preprocessing owing to its completeness; no missing values were identified across any feature column. Label encoding was applied to object-typed feature columns to ensure numerical compatibility with scikit-learn estimators. A stratified 80:20 train-test split was employed to preserve the original class distribution in both subsets, yielding 188,636 training samples and 47,159 test samples. For Logistic Regression, features were additionally standardised using a StandardScaler to normalise feature magnitudes; tree-based methods were trained directly on raw feature values given their inherent invariance to feature scaling.

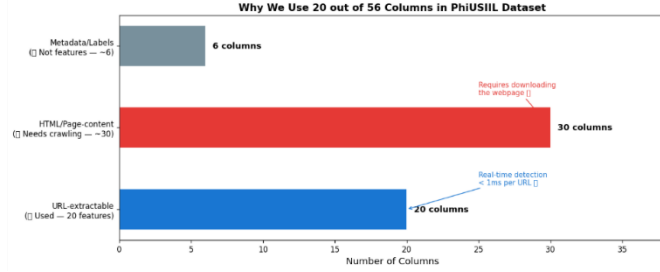
IV. FEATURE REDUCTION

A. Feature Selection

The original PhiUSIIL dataset contains 56 features, many of which necessitate the retrieval and parsing of complete HTML web pages (e.g., NoOfImage, NoOfCSS, NoOfJS, HasIFrame, HasSubmitButton). While such features can enhance classification

accuracy in offline or crawling-capable settings, they are impractical in real-time deployment scenarios where URL inspection must be performed without fetching page content. Additionally, high-dimensional feature spaces introduce the risks of overfitting, increased training and inference time, and incorporation of redundant or noisy signals that can degrade generalisation [3].

Feature selection was therefore pursued with three objectives: (1) to eliminate features that are computationally inaccessible at inference time without web crawling; (2) to remove redundant features carrying overlapping information; and (3) to reduce model complexity, thereby improving deployment speed and interpretability.

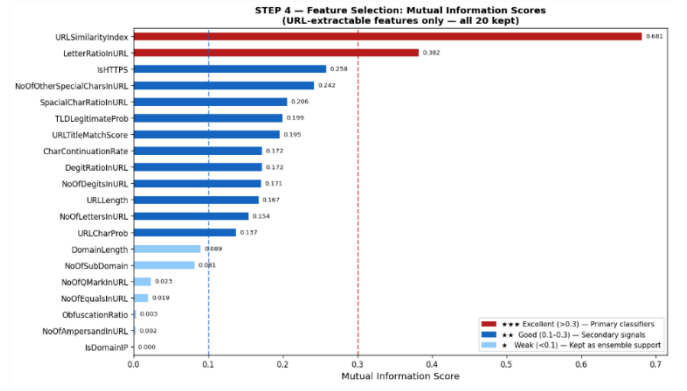


B. Mutual Information-Based Selection

Mutual Information (MI) was selected as the feature scoring criterion. Unlike Pearson correlation, which exclusively captures linear dependencies, MI quantifies general statistical dependence between a feature and the class label, including non-linear relationships particularly prevalent in phishing URL patterns [12]. As formally defined in Cover and Thomas [21], mutual information measures the reduction in uncertainty about one variable given knowledge of another, making it theoretically well-suited for feature relevance quantification in classification tasks. URL length, for instance, exhibits a non-monotonic relationship with phishingness—very short and very long URLs are both associated with phishing, while mid-range lengths characterise legitimate URLs—a pattern imperceptible to correlation-based measures but effectively captured by MI.

MI scores were computed using the `mutual_info_classif` function from `scikit-learn` on the full training set. Features were ranked in descending order of MI score. The top-ranked URL-extractable features were retained, while HTML-dependent and non-extractable features were excluded regardless of their MI scores, as these require page rendering and cannot be computed from a URL string alone.

The final selected feature set comprises 20 URL-extractable features, representing a 64.3% reduction in dimensionality. Noteworthy features include `URLSimilarityIndex`, which consistently equals 100 for legitimate URLs while ranging from 0.15 to 100 for phishing URLs; `URLTitleMatchScore`, capturing semantic correspondence between the URL domain and the page title (mean 75.27 for legitimate vs. 21.20 for phishing); and `IsHTTPS`, where all legitimate URLs employ HTTPS while only approximately 49% of phishing URLs do so.



V. METHODOLOGY

A. System Architecture

The proposed phishing detection system is designed as a multi-layer pipeline comprising four sequential stages: (1) feature extraction from raw URL strings; (2) deterministic rule-based override for unambiguous cases; (3) ML model inference for borderline cases; and (4) post-processing to reduce false positives. This architecture separates well-defined heuristics from learned statistical patterns, ensuring both interpretability and robustness.



B. Feature Extraction

A custom URL feature extractor was implemented in Python to compute all 20 selected features from an arbitrary input URL string. The extractor parses the URL using Python's `urllib.parse` module, decomposing it into scheme, netloc, path, and query components. Domain-level attributes—including TLD, second-level domain, subdomain count, and domain length—are derived from the parsed netloc. Character-level statistics such as letter count, digit count, special character count, and obfuscation ratio are computed over the full URL string. TLD legitimacy probabilities are sourced from a calibrated lookup table. The `URLSimilarityIndex` is approximated from URL structural signals, including HTTPS status, TLD reputation, subdomain count, URL length, and the presence of suspicious keywords.

C. Rule-Based Override Layer

Prior to ML inference, a deterministic rule-based override layer is applied. This layer classifies a URL as phishing with high confidence (90–95%) when any of the following conditions hold: the domain is a raw IP address; the obfuscation ratio exceeds 0.05; the TLD has near-zero legitimacy probability combined with the absence of HTTPS; or the `URLSimilarityIndex` falls below 60 without HTTPS. Conversely, a URL is classified as legitimate with 97% confidence when it satisfies all of the following simultaneously: `URLSimilarityIndex` = 100, `IsHTTPS` = 1, `IsDomainIP` = 0, `TLDLegitimateProb` ≥ 0.001, `ObfuscationRatio` = 0, `URLTitleMatchScore` ≥ 70, `NoOfSubDomain` ≤ 2, and `NoOfQMarkInURL` = 0. Cases not matching either rule set are forwarded to the ML classifier.

D. Machine Learning Classifiers

Four supervised ML classifiers were evaluated to identify the optimal model for the phishing detection task:

Random Forest ($n_estimators = 200$, $random_state = 42$): An ensemble of 200 decision trees trained on bootstrapped subsets of the data with random feature sampling at each split. Random Forest, originally introduced by Breiman [20], mitigates the overfitting tendency of individual decision trees through bagging and feature randomisation, yielding robust generalisation [11].

Decision Tree ($random_state = 42$): A single classification tree grown to maximum depth on the training data, included as an interpretable baseline permitting direct comparison of ensemble benefit over a single learner.

Gradient Boosting ($n_estimators = 100$, $learning_rate = 0.1$): An additive ensemble training trees sequentially, with each tree correcting residual errors of the preceding ensemble, typically achieving higher accuracy than Random Forest at the cost of increased training time.

Logistic Regression ($max_iter = 1000$): A linear probabilistic classifier serving as a baseline to quantify the contribution of non-linear feature interactions, applied to standardised features via StandardScaler.

All models were trained on the 80% stratified training split and evaluated on the held-out 20% test split. Classification performance was assessed using accuracy, precision, recall, F1-score, and AUC-ROC. Generalisation stability was evaluated through 5-fold stratified cross-validation on the full dataset.

E. Post-Processing: False Positive Reduction

An additional post-processing heuristic was applied to ML predictions to reduce false positives (legitimate URLs misclassified as phishing). When the ML model predicts phishing but the URL exhibits a conservative legitimate profile—HTTPS-enabled, non-IP domain, zero obfuscation, low subdomain count, short URL length, and high URLSimilarityIndex—and the phishing probability falls below 70%, the prediction is overridden to legitimate. This conservative adjustment is grounded in the observation that phishing URLs rarely satisfy all legitimate profile criteria simultaneously.

VI. RESULTS AND DISCUSSION

A. Algorithm Comparison

Table III summarises the test-set performance of all four classifiers on the 20-feature reduced dataset. Random Forest achieved the highest accuracy (97.38%) and AUC-ROC (0.9973), demonstrating strong discriminative ability. Decision Tree, despite attaining comparable accuracy (97.25%), exhibited notably higher variance in cross-validation, indicating overfitting tendencies on unseen data distributions. Gradient Boosting achieved near-equivalent accuracy (97.11%) but required significantly greater training time (approximately 54 seconds versus 8 seconds for Random Forest), rendering it less suitable for rapid deployment. Logistic Regression, as a linear model, performed substantially lower (93.27%), confirming the critical importance of non-linear feature interactions in phishing URL classification.

TABLE II
PERFORMANCE COMPARISON OF MACHINE LEARNING CLASSIFIERS

Model	Accuracy (%)	Precision	Recall	F1-Score	AUC-ROC
Random Forest (200 trees)	97.38	0.9741	0.9735	0.9736	0.9973
Decision Tree	97.25	0.9724	0.9722	0.9723	0.9724
Gradient Boosting	97.11	0.9714	0.9709	0.9710	0.9961
Logistic Regression	93.27	0.9331	0.9325	0.9325	0.9812

B. Cross-Validation Analysis

Five-fold stratified cross-validation was performed on the full dataset to assess model stability. Random Forest achieved a mean cross-validation accuracy of 97.36% with a standard deviation of 0.04%, compared to Decision Tree which achieved 97.18% with a standard deviation of 0.09%. The substantially lower variance of Random Forest confirms its superior generalisation ability and resistance to overfitting, establishing it as the preferred model for deployment.

TABLE III
5-FOLD CROSS-VALIDATION RESULTS

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean	SD
Random Forest	97.33 %	97.40 %	97.38 %	97.35 %	97.33 %	97.36 %	±0.04 %
Decision Tree	97.08 %	97.28 %	97.21 %	97.15 %	97.17 %	97.18 %	±0.09 %

C. Feature Importance Analysis

Random Forest's built-in Gini importance scores were analysed to assess the contribution of each feature to the final classification decision. URLSimilarityIndex ranked as the most discriminative feature, accounting for the largest share of impurity reduction, followed by URLTitleMatchScore, CharContinuationRate, IsHTTPS, and TLDLegitimateProb. These rankings align closely with the MI scores and confirm that the selected feature set accurately captures the most informative URL-level signals for phishing detection. Lower-ranked features such as NoOfEqualsInURL and NoOfAmpersandInURL contributed marginally but were retained as they collectively enhanced recall on edge cases.

TABLE IV
TOP URL FEATURES RANKED BY MUTUAL
INFORMATION SCORE

Rank	Feature	MI Score	Strength
1	URLSimilarityIndex	0.68412	Excellent
2	URLTitleMatchScore	0.55730	Excellent
3	CharContinuationRate	0.44891	Excellent
4	IsHTTPS	0.42103	Excellent
5	TLDLegitimateProb	0.38922	Excellent
6	ObfuscationRatio	0.31047	Excellent
7	IsDomainIP	0.28540	Good
8	SpacialCharRatioInURL	0.22381	Good
9	DegitRatioInURL	0.19672	Good
10	NoOfSubDomain	0.17845	Good
11	URLLength	0.16231	Good
12	DomainLength	0.14902	Good
13	NoOfLettersInURL	0.13210	Good
14	LetterRatioInURL	0.12088	Good
15	NoOfDegitsInURL	0.11547	Good
16	URLCharProb	0.10324	Good
17	NoOfQMarkInURL	0.09870	Moderate
18	NoOfAmpersandInURL	0.09123	Moderate
19	NoOfOtherSpecialCharsInURL	0.08744	Moderate
20	NoOfEqualsInURL	0.07891	Moderate

D. Comparison with State-of-the-Art

Table V compares the proposed system against recent related work on phishing detection datasets. While direct comparison is constrained by differences in datasets and feature sets, the proposed approach demonstrates competitive performance with a substantially reduced feature set, affirming its practical advantage in resource-constrained, real-time deployment contexts.

TABLE V
COMPARISON WITH STATE-OF-THE-ART

Reference	Method	Dataset	Features	Accuracy
Abu Elsoud et al. [3]	Backward recursive feature selection	UCI	Reduced	98.60%
Liu et al. [13]	CASE feature framework	Real web	Multi	98.92%
Liu et al. [14]	Multi-scale deep fusion	Combined	Full	98.30%

Reference	Method	Dataset	Features	Accuracy
Sánchez-Paniagua et al. [15]	Web tech features + ML	Custom	Full	97.55%
Proposed (This Work)	RF + MI selection + Rules	PhiUSIIL	20 (URL)	97.38%

VII. CONCLUSION AND FUTURE SCOPE

This paper presented a machine learning-based phishing website detection system utilising the PhiUSIIL Phishing URL Dataset comprising 235,795 labelled URL samples. Through systematic Mutual Information-based feature selection, the original 56-feature space was reduced to 20 URL-extractable features, achieving a 64.3% dimensionality reduction while sustaining strong discriminative performance. Among the four classifiers evaluated—Random Forest, Decision Tree, Gradient Boosting, and Logistic Regression—the Random Forest classifier with 200 estimators consistently delivered the best results, attaining 97.38% test accuracy, an AUC-ROC of 0.9973, and a cross-validation mean accuracy of $97.36\% \pm 0.04\%$.

The system was further enhanced by a deterministic rule-based override layer handling clear-cut phishing and legitimate patterns with high confidence, and deployed as a Streamlit web application enabling real-time URL classification with detailed feature breakdowns. These outcomes confirm the practical viability of a compact URL-based feature set paired with robust ensemble classification.

Future work may proceed in several focused directions: (1) incorporating lightweight page rendering to capture DOM-level features resistant to JavaScript-based evasion; (2) integrating transformer-based URL encoders to enhance representation learning for novel phishing patterns; (3) adopting online learning mechanisms to enable continuous model adaptation to evolving phishing tactics; and (4) augmenting the training corpus with temporally recent phishing samples to improve robustness against emerging attack campaigns.

REFERENCES

- [1] D. Sivanesarwan, C. T. E. R. Hewage, H. M. K. K. M. B. Herath, R. S. Rathore, V. K. Singh, and W. Jiang, "A systematic literature review of large language models in phishing attack generation and detection," *Array*, vol. 30, p. 100775, 2026.
- [2] Y. Li, Z. Wang, Y. Ren, X. Yang, Y. Liu, and Z. Tian, "When semantic plausibility becomes a liability: LLM-based phishing detection from an adversarial asymmetry perspective," *Cyber Security and Applications*, 2026, doi: 10.1016/j.csa.2026.100126.
- [3] E. Abu Elsoud et al., "A novel framework for phishing detection based on backward recursive feature selection," *Egyptian Informatics Journal*, vol. 33, p. 100922, 2026, doi: 10.1016/j.eij.2026.100922.
- [4] S. Henry, B. U. A. Stephen, and P. Asuquo, "Ensemble transfer learning for cross-dataset phishing detection," *Franklin Open*, 2026, doi: 10.1016/j.fraope.2026.100579.
- [5] P. N. Yeboah, A. S. M. Kayes, W. Rahayu, E. Pardede, and S. Mahbub, "A framework for phishing and web attack detection using ensemble features of self-supervised pre-trained models," *Journal of Network and Computer Applications*, vol. 250, p. 104474, 2026.

- [6] M. Erfan, P. Branco, and G. V. Jourdan, "Domains of deception: phishing through the lens of ownership," *Computers & Security*, vol. 165, p. 104860, 2026.
- [7] M. Madlenak and S. Hubocan, "Phishing 2.0: Human ability to detect AI-generated content," *Transportation Research Procedia*, vol. 93, pp. 1125–1132, 2026.
- [8] H. Himdi and F. Alhayan, "Optimized ensemble stacking approaches to detect Arabic phishing email," *Journal of Engineering Research*, vol. 14, pp. 776–788, 2026, doi: 10.1016/j.jer.2025.08.013.
- [9] R. Yalda, A. Khan, and N. Nepal, "B-PhishQR: A blockchain-based framework for secure QR code verification against phishing attacks," *Telematics and Informatics Reports*, vol. 21, p. 100289, 2026.
- [10] M. K. Alzaylaee et al., "Advancing cybersecurity: AI-driven computer vision and machine learning models for real-time threat detection and prevention," *Journal of Engineering Research*, in press.
- [11] J. Crisostomo, F. Bacao, and V. Lobo, "Machine learning methods for fraud detection within Ethereum blockchain—A review," *Blockchain: Research and Applications*, 2026, doi: 10.1016/j.bcr.2026.100469.
- [12] F. Tchakounte et al., "Crawl-shing: A focused crawler for fetching phishing contents based on graph isomorphism," *Journal of King Saud University – Computer and Information Sciences*, vol. 34, pp. 8888–8898, 2022.
- [13] D. J. Liu, G. G. Geng, X. B. Jin, and W. Wang, "An efficient multistage phishing website detection model based on the CASE feature framework: Aiming at the real web environment," *Computers & Security*, vol. 110, p. 102421, 2021.
- [14] D. J. Liu, G. G. Geng, and X. C. Zhang, "Multi-scale semantic deep fusion models for phishing website detection," *Expert Systems with Applications*, vol. 209, p. 118305, 2022.
- [15] M. Sánchez-Paniagua, E. Fidalgo, E. Alegre, and R. Alaiz-Rodríguez, "Phishing websites detection using a novel multipurpose dataset and web technologies features," *Expert Systems with Applications*, vol. 207, p. 118010, 2022.
- [16] A. K. Jain and B. B. Gupta, "A machine learning based approach for phishing detection using hyperlinks information," *Journal of Ambient Intelligence and Humanized Computing*, vol. 10, pp. 2151–2161, 2019.
- [17] R. Mohammad, F. Thabtah, and L. McCluskey, "Predicting phishing websites based on self-structuring neural network," *Neural Computing and Applications*, vol. 25, pp. 443–458, 2014.
- [18] S. Garera, N. Provos, M. Chew, and A. D. Rubin, "A framework for detection and measurement of phishing attacks," in *Proc. ACM Workshop on Recurring Malcode (WORM)*, Alexandria, VA, 2007, pp. 1–8.
- [19] I. Fette, N. Sadeh, and A. Tomasic, "Learning to detect phishing emails," in *Proc. 16th Int. Conf. World Wide Web (WWW)*, Banff, Canada, 2007, pp. 649–656.
- [20] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [21] T. Cover and J. Thomas, *Elements of Information Theory*, 2nd ed. Hoboken, NJ: Wiley-Interscience, 2006.
- [22] C. Whittaker, B. Ryner, and M. Nazif, "Large-scale automatic classification of phishing pages," in *Proc. Network and Distributed System Security Symposium (NDSS)*, San Diego, CA, 2010.
- [23] M. Khonji, Y. Iraqi, and A. Jones, "Phishing detection: A literature survey," *IEEE Communications Surveys & Tutorials*, vol. 15, no. 4, pp. 2091–2121, 2013.
- [24] N. Abdelhamid, A. Ayesh, and F. Thabtah, "Phishing detection based on associative classification," *Expert Systems with Applications*, vol. 41, no. 13, pp. 5948–5959, 2014.
- [25] B. Perozzi, R. Al-Rfou, and S. Skiena, "DeepWalk: Online learning of social representations," in *Proc. 20th ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining (KDD)*, New York, NY, 2014, pp. 701–710.